

Daehyun Choi

Postdoctoral Researcher · Fluid Mechanics · Bio-inspired Propulsion · Physical AI

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Summary

I am a postdoctoral researcher at the University of Colorado Boulder, applying fluid mechanics to vision AI, robotics, digital twins, bioinspiration, and animal behavior. My current research focuses on data-driven shape optimization of physical AI and soft robotic systems.

Highlights

- **First-authored articles:** *J. Fluid Mech.* (×3, JFM Emerging Scholar Best Paper Prize), *Phys. Fluids* (Editor's Pick), *Sci. Rep.*, *Integr. Comp. Biol.*; >3 papers under review and in preparation (arXiv preprints available).
- **Awards & Grants** (~\$700k total): DARPA Young Faculty Award (\$500k, technical lead); Sejong International Postdoctoral Fellowship, NRF Korea (\$140k); Domestic Postdoctoral Scholarship, NRF Korea (\$50k); NEREM Travel Grant; IUTAM Travel Grant.

Research Interests

- **Fluid mechanics · Fluid-structure interaction & vortex dynamics · Multiphase flows · Bioinspired propulsion & locomotion · Vision AI · Multi-fidelity & multi-objective data-driven optimization · Soft robotics · Physical AI**

Skills

- **Hardware:** End-to-end device prototyping; high-speed imaging, 2D/3D PIV, 3D DIC, stereo-camera reconstruction, dynamometry; wind tunnel, water channel, pressure chamber, large-scale water tank experiments; PCB design & fabrication, circuit design, BLE wireless sensing, microcontroller data acquisition; silicone elastomer fabrication, micro-robot fabrication; biological fieldwork.
- **Software:** C++, Python, MATLAB (>10 years); AI-assisted development (expert level); CNN-based vision AI — DeepBubbleVelocimetry [🔗★9] (developed); co-developed fluid-structure interaction simulation (OpenFOAM + CalculiX + preCICE) [arXiv]; Bayesian/multi-objective optimization (BoTorch); multiphysics simulation (ANSYS Fluent, MATLAB).
- **Leadership:** DARPA team lead, coordinating CFD (Univ. of Birmingham), experiment & optimization (Georgia Tech), and DARPA program manager; mentored >12 students (graduate, undergraduate, high school); peer reviewer: *J. Fluid Mech.*, *Phys. Fluids*; press: The Conversation, PBS NewsHour, Science News, & Georgia Tech News; proposal writing & grant acquisition (~\$700k); international collaboration (Korea–USA–Brunei); MSCA Seal of Excellence (96.6/100, EF-ENV).

Academic Employment

University of Colorado Boulder

Postdoctoral Researcher

Boulder, CO, USA

2026 – present

Georgia Tech

Postdoctoral Researcher

Atlanta, GA, USA

2023 – 2026

Ajou University

Postdoctoral Researcher

Suwon, Korea

2023

Education

Seoul National University
M.S. & Ph.D. in Mechanical Engineering

Seoul, Korea
2022

Seoul National University
B.S. in Mechanical Engineering

Seoul, Korea
2015

Publications

Bold denotes the author (Daehyun Choi). * denotes co-first authorship. † denotes co-corresponding authorship.

Journal Articles

1. **Daehyun Choi**, Kai Lauren Yung, Halley Wallace, Jacob Harrison, & M. Saad Bhamla, “Interfacial dynamics on the water-hopping behavior of mudskipper,” *In preparation*.
2. **Daehyun Choi**, Paras Singh, Halley Wallace, Chandan Bose, & M. Saad Bhamla, “Deep-learning Bayesian surrogate model for thrust enhancement of soft nozzle,” *In preparation*.
3. **Daehyun Choi**, Paras Singh, Ian Bergerson, Minho Kim, Jieun Park, Halley J. Wallace, Kenny Zhang, Sandy Y. Hsieh, Aqua T. Asberry, Theodore A. Uyeno, William F. Gilly, Hyungmin Park, Daeshik Kang, Chandan Bose, & M. Saad Bhamla, “Squid-inspired soft superpropulsion,” *arXiv:2605.03239* (2026). *Under review*. [arXiv](#)
4. **Daehyun Choi**^{*}, Paras Singh^{*}, M. Saad Bhamla, & Chandan Bose, “Computational and reduced-order modelling of elastic wave-driven impulse enhancement in pulsed jets through passively flexible nozzles,” *arXiv:2605.17319* (2026). *Under review*. [arXiv](#)
5. Paras Singh, Yakesh Karki, Victor Hernandez, **Daehyun Choi**, M. Saad Bhamla, & Chandan Bose, “Multi-objective Bayesian optimization of rigid and flexible nozzles for energy-efficient pulsed jet propulsion,” *arXiv:2606.19252* (2026). *Under review*. [arXiv](#)
6. **Daehyun Choi**^{*†}, Ian Bergerson^{*}, Hengjia Zhu^{*}, Tianjun Lan, & M. Saad Bhamla[†], “Swim-and-breach at palm scale: a rudder-steered two-propeller underwater robot platform with differential-thrust pitch control,” *arXiv:2609.17240* (2026). *Under review*. [arXiv](#)
7. Pankaj Rohilla, Johnathan Nathaniel O’Neil, Chandan Bose, Victor M. Ortega-Jimenez, **Daehyun Choi**, & M. Saad Bhamla, “Epineuston vortex recapture enhances thrust in tiny water skaters,” *Under review*. [PDF](#) [bioRxiv](#)
8. **Daehyun Choi**, Kai Yung, Ian Bergerson, Halley Wallace, Ulmar Grafe, & M. Saad Bhamla, “Three-dimensional tracking method for water-hopping mudskippers in natural habitats,” *Integr. Comp. Biol.*, icaf139 (2025). [PDF](#) [DOI](#)
9. Pankaj Rohilla^{*}, **Daehyun Choi**^{*}, Halley Wallace, Kai Lauren Yung, Juhi Deora, Atharva Lele, & M. Saad Bhamla (*equal contribution), “Mastering the Manu – How humans create large splashes,” *J. R. Soc. Interface*, 15, 2 (2025). [PDF](#) [DOI](#) [The Conversation](#)
10. **Daehyun Choi** & Hyungmin Park, “Enhanced impulse and entrainments of the pulsed circular jet by flexible nozzle,” *J. Fluid Mech.*, 996, A6 (2024). [PDF](#) [DOI](#)
11. **Daehyun Choi**^{*}, Hyunseok Kim^{*}, & Hyungmin Park (*equal contribution), “Bubble velocimetry using the conventional and CNN-based optical flow algorithms,” *Sci. Rep.*, 12, 11879 (2022). [PDF](#) [DOI](#) [Code](#)
12. **Daehyun Choi** & Hyungmin Park, “Flow-structure interaction of the starting jet through the flexible circular nozzle,” *J. Fluid Mech.*, 949, A39 (2022). [PDF](#) [DOI](#)
13. **Daehyun Choi**, Jungwon Byun, & Hyungmin Park, “Analysis of the liquid column atomization by annular dual-nozzle gas jet flow,” *J. Fluid Mech.*, 943, A25 (2022). [PDF](#) [DOI](#)
14. Dokeun Lee, **Daehyun Choi**, Hyungmin Park, & Sungjae Kim, “Electroconvective circulating flows by asymmetric coulombic force distribution in multiscale porous membrane,” *J. Membr. Sci.*, 636, 119286 (2021). [PDF](#) [DOI](#)

15. **Daehyun Choi** & Hyungmin Park, “Flow around in-line sphere array at moderate Reynolds number,” *Phys. Fluids (Editor’s Pick)*, 30, 097104 (2018). [PDF](#) [DOI](#)

Conferences & Seminars

1. “Hydrodynamics of water-hopping behavior of mudskippers,” 2026 *Society for Integrative and Comparative Biology (SICB)*, Oral presentation (Symposium). **USA**, 2026.
2. “Squid-inspired flexible nozzles improve thrust in rotary propulsors using a multi-fidelity approach,” 2026 *Society for Integrative and Comparative Biology (SICB)*, Oral presentation. **USA**, 2026.
3. “Squid funnel-inspired passively flexible nozzles for underwater propulsion,” 2026 *Society for Integrative and Comparative Biology (SICB)*, Oral presentation. **USA**, 2026.
4. “Geometrical effects of free-end nozzles on wave reflection in self-oscillating flexible nozzles,” 78th *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Houston, USA**, 2025.
5. “Bio-inspired flexible nozzle design for enhanced underwater propulsion,” *Fluid and Biodynamics Seminar, Max Planck Institute for Dynamics and Self-Organization*, Oral presentation (Invited). **Göttingen, Germany**, 2025.
6. “Squid-inspired nozzles for enhanced thrust in rotary propulsors,” 2025 *European Mechanics Society: Bio-inspired Fluid-Structure Interaction (EUROMECH)*, Oral presentation. **Vienna, Austria**, 2025.
7. “Squid-inspired nozzles for enhanced thrust in rotary propulsors,” 2025 *American Physical Society March Meeting*, Oral presentation. **Anaheim, USA**, 2025.
8. “Non-invasive 3D tracking of water-hopping mudskippers in natural habitats,” 2025 *Society for Integrative and Comparative Biology (SICB)*, Poster presentation. **Atlanta, USA**, 2025.
9. “Physical modelling of water-hopping mudskipper,” 77th *Annual Meeting of the APS Division of Fluid Dynamics*, Poster presentation. **Salt Lake City, USA**, 2024.
10. “Water-hopping behavior of mudskipper,” 2024 *IUTAM Symposium on Capillarity and Elastocapillarity in Biology*, Poster presentation. **Seoul, Korea**, 2024.
11. “Hydrodynamics of water-hopping mudskipper,” 2024 *International Physics of Living Systems Annual Meeting*, Oral presentation. **Trieste, Italy**, 2024.
12. “Hydrodynamic study on the water-hopping behavior of mudskipper,” 2024 *American Physical Society March Meeting*, Oral presentation. **Minneapolis, USA**, 2024.
13. “Vortex mechanism in microvelia-inspired water-walking for thrust optimization using robotic design,” 2024 *Society for Integrative and Comparative Biology (SICB)*, Poster presentation. **Seattle, USA**, 2024.
14. “Experimental study on the starting jet through the flexible circular nozzle,” 2022 *World Congress on Advances in Civil, Environmental and Materials Research (ACEM)*, Oral presentation. **Seoul, Korea**, 2022.
15. “On the optimal flexibility condition of an impulsive starting jet nozzle for thrust generation,” 73rd *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Online**, 2020.
16. “Experimental study on the effect of nozzle flexibility on the evolution of a starting liquid jet,” 72nd *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Seattle, USA**, 2019.
17. “Study on dual-nozzle jet flows for the application of liquid atomization,” 71st *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Atlanta, USA**, 2018.
18. “Experimental study on the flow around in-line array of spheres,” 69th *Annual Meeting of the APS Division of Fluid Dynamics*, Oral presentation. **Portland, USA**, 2016.

Research Project Experience

Squid-Inspired Nozzles for Enhanced Efficiency and Thrust in Rotary Propulsors
Supported by *Defense Advanced Research Projects Agency*

Georgia Tech
2024 – present

Hydrodynamics of Water-Hopping Mudskipper and Biomimetic Robot Development Supported by <i>National Research Foundation of Korea – Sejong International Postdoctoral Fellowship</i>	Georgia Tech 2023 – present
Development of Cephalopod-Inspired Jet-Propelled Soft Underwater Vehicle Supported by <i>National Research Foundation of Korea – Domestic Postdoctoral Fellowship</i>	Ajou University 2022 – 2023
Physical-Based Model for Film Thickness and Spray Characteristics Depending on Operation Conditions Supported by <i>Daewoo Shipbuilding & Marine Engineering (DSME)</i>	Seoul National University 2021 – 2022
Apparatus for Determining Removal of Foley Catheter Supported by <i>Yonsei Severance Hospital</i>	Seoul National University 2020 – 2022
Measurement of Bubble and Particle in Liquid Supported by <i>Samsung Electro-Mechanics</i>	Seoul National University 2018 – 2019
Analysis of Molten Metal Atomization Using Multi-physics Simulation Supported by <i>Samsung Electronics</i>	Seoul National University 2018 – 2018
Effects of Superhydrophobic Surface on the Wake of Circular Cylinder Supported by <i>Samsung Advanced Institute of Technology (SAIT)</i>	Seoul National University 2016 – 2016

Awards & Grants

- 2025 **Emerging Scholar Best Paper Prize (Top 10)**, *Journal of Fluid Mechanics*, **Cambridge University Press**
- 2024 **Technical Leader – DARPA Young Faculty Award (\$500k)**, **Defense Advanced Research Projects Agency**
- 2024 **Robert M. Nerem International Travel Award (\$3k)**, **Georgia Tech**
- 2024 **Young Researcher Travel Award (\$1k)**, IUTAM Symposium on Capillarity and Elastocapillarity in Biology
- 2023 **Sejong International Postdoctoral Fellowship (\$140k)**, **National Research Foundation of Korea**
- 2022 **Best Thesis Award**, Dept. of Mechanical Engineering, **Seoul National University**
- 2022 **Domestic Postdoctoral Scholarship (\$50k)**, **National Research Foundation of Korea**
- 2016 **BK21+ Doctoral Scholarship (\$10k)**, Ministry of Education, Korea
- 2015 **Presidential Scholarship for Undergraduates (\$30k)**, Korea
- 2014 **1st Prize – EDISON Challenge (CFD contest, \$1k)**, Ministry of Science, ICT and Future Planning, Korea

Mentoring & Teaching

- 2023 – **Research Mentor**, Graduate (3) and undergraduate (5) students; 1 high school student, Georgia Tech
- 2022 – **Research Mentor**, 2 graduate students, Ajou University
- 2016 – **Undergraduate Thesis Advisor**, 3 undergraduate students, Seoul National University
- 2016 – **Teaching Assistant**, Mechanical Engineering Laboratory: Flow Measurements, Seoul National University
- 2017

Patents (Lead Inventor)

- 2026 **US Patent Application, No. 19/568,682**, Compliant Nozzle System and Method for Rotary Propulsion, Georgia Tech Research Corporation
- 2026 **US Patent Application, No. 19/568,695**, Compliant Nozzle System and Method for Fluid Jet Impulse, Georgia Tech Research Corporation
- 2021 **KR Patent, 10-2021-0149171**, System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with flow and hydraulic pressure, Korea
- 2021 **KR Patent, 10-2021-0149176**, System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with hydraulic pressure and UASIF, Korea
- 2021 **KR Patent, 10-2021-0149175**, System for reducing insertion pressure of the ureteral access sheath for RIRS by artificially inducing hydronephrosis and measuring induced hydronephrosis with UASIF, Korea
- 2020 **PCT Patent, PCT/KR2020/018848**, Apparatus for determining removal of Foley catheter, International
- 2020 **KR Granted Patent, 10-2020-0064729**, System for reducing ureteral insertion pressure by artificially triggering hydronephrosis based on measurement of pressure associated with the insertion of the ureteral access sheath for RIRS, Korea
- 2020 **KR Granted Patent, 10-2020-0064731**, System for reducing ureteral entry pressure by generating vibrations in the guide wire for the insertion of the ureteral access sheath for RIRS, Korea
- 2020 **KR Granted Patent, 10-2020-0000288**, Apparatus for determining removal of Foley catheter, Korea